

Dominic Lemoine de Martigny

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Education

University of Waterloo

Bachelor of Software Engineering - Co-op, GPA: 3.9/4.0

Waterloo, ON

Sep. 2024 - Apr. 2028

Skills

Languages: Python, TypeScript, C, C++, SQL, JavaScript, HTML, CSS, Java, C#, Ruby

Technology: React, Node.js, AWS (Lambda, EC2, S3, CloudFormation, RDS), MySQL, PostgreSQL, Supabase, Firebase, GraphQL, Next.js, Vite, LangChain, Tailwind, Flask, Flutter, PyTorch, OpenCV, Django, FastAPI, Spring Boot, ASP.NET

Development: Docker, Git, Kubernetes, Postman, CI/CD, Playwright, GTest, Pytest, Jest, Linux, Colab, Jupyter

Experience

Software Engineer Intern

Versaterm

Jan. 2026 - Apr. 2026

Vancouver, BC

- Incoming **Winter 2026!** Building software for public safety agencies using **Java**, **Spring Boot**, **C#**, and **ASP.NET**.

Software Engineer Intern

ALS Geoanalytics

May. 2025 - Aug. 2025

Montr  al, QC

- Deployed custom CV segmentation models using **Kubernetes** and **EC2**, cutting labeling time from ~**5 min** to **10 s**.
- Developed an **S3** cloud annotation system handling **5TB+** of data, enabling session persistence for geoscientists.
- Engineered a microservices architecture with **FastAPI**, implementing a **RAG**-powered chatbot using **LangChain** and **FAISS** to query thousands of internal research documents, reducing retrieval time by up to ~**40%**.
- Built a **React/Django** platform integrating **AWS Lambda** and **RDS** to drive scalable, self-service custom model training, featuring real-time **WebSocket** dashboard updates and **Amplify** authentication.
- Implemented a comprehensive **E2E** testing suite across production with **GitHub Actions** and **Playwright**.

Software Developer

GCE Global Solutions

Jun. 2024 - Sep. 2024

Montr  al, QC

- Overhauled a legacy PHP website into a modern responsive **React + Vite** SPA, improving page load times by ~**60%**.
- Automated legal entity extraction with a Python + **spaCy** NLP pipeline, reaching **92%** accuracy on **500+** contracts.
- Built a **PostgreSQL** database and **GraphQL** API for **10K+** legal records to streamline employee case research.

Projects

uwGuessr.com | *TypeScript, Next.js, GraphQL, Supabase, AWS (Lambda, S3), Tailwind, Mapbox, Auth0*

- Launched **uwGuessr**, a Waterloo geoguessing web game, reaching **110,000+** visits and **9,500+** unique users.
- Implemented a real-time leaderboard system using **GraphQL** subscriptions and **Supabase** PostgreSQL triggers, processing **200+** daily submissions with sub-**300ms** response times
- Built a user-submitted image upload workflow with client-side compression, **Lambda** processing, and **S3** storage.
- Deployed security features including **rate limiting**, **proxy detection**, and **Auth0** to counter active DDOS attacks.

GLASSES | *Python, Flask, React Native, JavaScript, Git, Raspberry Pi*

- Designed and developed **real-time** music recognition glasses using a Raspberry Pi and integrated display, achieving **95%** accuracy in song identification with Shazam API within **3 seconds** via audio fingerprinting.
- Built a **Flask** backend to process streamed audio chunks, displaying synced lyrics to the lens with **<200ms** latency.
- Collaborated in a team to seamlessly integrate a **React Native** app, Raspberry Pi, display module, and server.

RememberU (Hack the North) | *Flutter, Firebase, Python, Flask, OpenCV, Gemini, Raspberry Pi, Postman*

- Built a wearable **AI-powered** device and **Flutter** app with a **Flask + Firebase** backend, combining **lip-reading** and **facial recognition** to capture conversations and deliver personalized memory recall during busy events.
- Integrated **Gemini**, **OpenCV**, and lip-reading APIs to generate real-time, **context-aware** conversation summaries.